LESSON GUIDE

Technology Plus

Unit 3: Hardware

1.3.2 – Transfer Rates

Lesson Overview:

In this lesson, students will explore throughput and how it relates to networking, differentiate between bits and bytes, differentiate between throughput and bandwidth, identify obstacles of throughput, and compare common throughput units.

Students will:

- Identify and define throughput and its obstacles.
- Differentiate between bits and bytes.
- Differentiate between bandwidth and throughput.
- Compare common throughput units.
- Apply knowledge of throughput units to real-world computing scenarios.

Guiding Question: How do different throughput units measure the speed of data transfer and how does it impact everyday activities?

Suggested Grade Levels: 8-12

Technology Needed: No

Approximate Time Required: 40-50 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 1.3 – Compare and Contrast common units of measure.
  - Throughput unit
    - Bits per second (bps)
    - Kilobits per second (Kbps)
    - Megabits per second (Mbps)
    - Gigabits per second (Gbps)
    - Terabytes per second (Tbps)

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Lesson Background

Background Information:

Throughput is a fundamental concept in networking and data communications that refers to the actual rate at which data is successfully transmitted over a communication channel. It is expressed in bits per second (bps) and its larger multiples – kilobits per second (Kbps), megabits per second (Mbps), gigabits per second (Gbps), and terabits per second (Tbps). The jump between each unit is by a factor of 1,000, and not 1,024 as in file sizes. Throughput provides a realistic measure of network performance because it accounts for real-world factors such as latency, transmission errors, protocols, bandwidth limit, and network traffic.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 1.3.2 – Transfer Rates • Activity Slides 1.3.2 – Data Speed Detective
Materials per Student	• Student Notes 1.3.2 – Transfer Rates • Student Handout 1.3.2 – Transfer Rates • Assessment 1.3.2 – Transfer Rates (optional)

Cyber Terms:

Term	Definition
throughput	the rate data moves from one place to another over a network in one second, measured by bits per second
bit (b)	short for "binary digit", represented as a 0 or 1 and comprises the most basic unit of information storage in computing
bandwidth	the maximum amount of data that can be transmitted over a network connection, measured by bits per second

1.3.2 – Transfer Rates

Guiding Question: How do different throughput units measure the speed of data transfer and how does it impact everyday activities?

Lesson Launch (5 minutes)

- Imagine you are filling a water bottle from a faucet. The water flowing out fills the bottle faster or slower depending on how much water comes through per second.
- The faster the water flows, the quicker your water bottle fills. If the faucet is dripping slowly, it takes a long time to fill the bottle.
- In computer networks, data flows through cables or wireless signals. Throughput, like that flow, is the actual rate at which data moves from one point to another over a network.

The Speed of Data: Throughput in Action (15-20 minutes)

- Distribute the Student Handout 1.3.2 – Transfer Rates and have students fill in the answers during the lesson.
- Introduce ***throughput*** regarding networking and its importance.
    - Throughput measures how much data moves from one place to another in a given block of time. While the concept applies to many areas (factories, pipelines, etc.), in networking it means how much data successfully transfers per second.
- Discuss the difference between ***bits*** and bytes.
    - Emphasize that network speeds are measured in **bits per second**, while file sizes are measured in bytes. Highlight the difference in jumps: 1,000 for bits (Kbps → Mbps → Gbps), 1,024 for bytes (KB → MB → GB).
    - Students may confuse it with bytes, which is a 1,024 jump – used when counting file sizes.
- Discuss the difference between ***bandwidth*** and throughput.
    - Clarify that bandwidth is the **maximum possible capacity** of a connection, while throughput is the **actual speed** achieved. Explain that obstacles like latency, congestion, or errors can reduce throughput below the available bandwidth.
    - Discuss the obstacles that cause lower throughput.
- Introduce each throughput unit, its capacity, usage, and examples.
    - Bits per second (bps): like sending one word at a time.
    - Kilobits per second (Kbps): like sending a letter by mail.
    - Megabits per second (Mbps): like sending a box by courier.
    - Gigabits per second (Gbps): like sending cargo by plane.
    - Terabits per second (Tbps): like sending multiple cargo planes per second.

Activity: Data Speed Detective (15-20 minutes)

- Display Activity Slides 1.3.2 – Transfer Rates, have students record their answers in the handout.
- Choose if you want students to work independently or in a small group.
- Read each scenario aloud or let students read independently or in their group.
- Students will analyze the scenario and determine which throughput unit best matches the situation, then explain why.
- Provide students with time to write their answers.

Scenarios and Answers

- A home Wi-Fi network streams multiple 4K videos at the same time without buffering. The connection needs to handle large amounts of data quickly for smooth playback.

  Gbps, because gigabit speeds support multiple simultaneous 4K streams without buffering.

- A large tech company transfers massive amounts of data between data centers located in different countries. The connection must be extremely fast to keep data synchronized in real time.

  Tbps, because this speed is necessary to efficiently handle enormous volumes of data moving between global data centers.

- An old dial-up internet connection from the early 2000s downloads text files very slowly. It cannot support video streaming or large file downloads.

  Kbps, because dial-up connections were measured in kilobits per second and are too slow for high-data activities like video streaming.

- A small business regularly uploads documents and presentations to cloud storage. Their internet connection is fast enough to complete uploads in a reasonable time but isn't extremely fast.

  Mbps, because megabit speeds are sufficient for moderate tasks like uploading documents and presentations without major delays.

- A gamer competes in an online tournament where real-time response is critical. Their internet connection must quickly send and receive data to avoid lag.

  Gbps, because gigabit speeds provide the fast data transfer capacity needed for competitive gaming and help minimize lag when combined with low latency.

- A person streams a music playlist on their smartphone while on the go. The data usage per song is low, but the connection should be stable.

  Mbps, because streaming music requires relatively low-to-moderate bandwidth, which megabit speeds can easily handle.

- An internet backbone connects major cities and carries data traffic for millions of users. The network must handle huge amounts of simultaneous data transfers.

  Tbps, because this speed is required for backbone networks to support large-scale, high-volume data transfers between cities.

- Sending text messages over a slow mobile connection in a rural area takes a few seconds to arrive. The data packets are small and don't need much bandwidth.

    Kbps, because text messaging uses very little data and can work on low-speed connections.

Putting It All Together (5 minutes)

- Review what students learned today by discussing the questions below as a class?
- What is throughput and why is it important?

    Throughput measures how much data moves from one place to another in one second. It is important because it determines how fast you can download, stream, or upload without delays. Throughput also affects how smoothly apps, websites, and online services work and is a key performance factor for internet service providers, cloud services, data centers, and home networks.

- What is bandwidth, and how does it impact throughput?

    Bandwidth is the maximum possible data rate of a connection. Throughput cannot be higher than the available bandwidth, but it is often lower due to obstacles such as latency or congestion.

- What obstacles can affect throughput?

    Throughput can be reduced by latency (delays in sending and receiving data), errors (when corrupted packets must be resent), protocols (overhead from rules for data transfer), limits set by the internet service provider, and network traffic or congestion.

- What are the units of throughput?

    The units of throughput are bits per second (bps), kilobits per second (Kbps), megabits per second (Mbps), gigabits per second (Gbps), and terabits per second (Tbps).